Seminar for Master Students in Software Engineering

Master Thesis Proposal Draft

Design and Evaluation of a Hybrid Graph–Table Interface with Embedded Recommender System for Study Planning

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1 Problem Statement and Motivation

Students in higher education face increasing challenges when planning their studies. Curricula often involve complex prerequisites, elective choices, and institutional rules that are difficult to navigate with static study guides or standardized plans [20, 17]. Individual circumstances, such as failed courses, varying study speeds, or limited access to advisors, further complicate planning and may negatively affect performance, graduation times, and motivation [26, 9]. Existing approaches address these challenges but continue to face limitations. Graph-based systems often focus narrowly on visualizing course dependencies [2, 19, 22], while table-based assistants center around semester-level planning [13, 14, 5]. This leaves a gap: integrating a graph and a table into a single student-planning workflow so students are well informed about course dependencies in their curricular (graph) and can manage their schedule and workload (table). Beyond these structural aids, curriculum analytics approaches provide useful insights at an institutional level but rely heavily on data of Learning Management Systems (LMS) or Content Management Systems (CMS), which limit their transferability across institutions [20, 12, 1, 8, 15]. More adaptive solutions, such as Educational Recommender Systems (ERS), demonstrate strong algorithmic potential [7, 26, 11, 21, 3], but to the best of our knowledge, they have not yet been embedded into student-facing planning environments. This raises the question of to what extent these ERS can actually support students in their planning workflow. Building on these challenges, we formulate the following main research question:

Main Research Question How can a hybrid graph–table planner, augmented with recommendations learned from students' interaction data, be designed and implemented to support study planning and enhance user experience?

2 Aim of the Thesis

This thesis proposes a unified approach to study planning: a hybrid graph–table interface that informs about course dependencies and enables semester scheduling, enriched by recommendations learned from students' own interactions. The recommender system is not the main focus of this thesis, but serves as a supporting component to demonstrate basic recommendation techniques within the planning interface. We aim to understand how graph- and table-based visualization of curriculum structures can be jointly orchestrated within authentic planning workflows, derive and embed data-driven recommendations that fit these workflows, and assess the resulting system's user experience. Therefore we divide our main research question into four sub-questions:

Sub-Question 1 How can we design a hybrid interface that combines graph-based visualization with table-based planning to help students plan their studies and understand course dependencies?

Sub-Question 2 How can data generated through students' interactions with the interface be used to develop a recommender system?

Sub-Question 3 How can the generated recommendations be effectively integrated into the hybrid graph–table interface and within a student planning workflow?

Sub-Question 4 How do students experience the hybrid interface and its embedded recommendations in terms of user experience (effectiveness, efficiency, and satisfaction)?

3 Methodology and Expected Results

To answer the four sub-questions introduced in Section 2, our tool addresses them methodologically as follows:

Hybrid Interface (graph, table) We investigate how a dependency graph and a semester-planning table can be combined, and used in tandem, within a coherent student planning workflow. First, we will analyze students' planning workflows and needs to identify where a graph's structural overview and a table's scheduling capabilities add the most value. To inform this analysis, we will conduct a small user study to gather user insights and needs. Based on these, we will prototype coordination mechanisms that link both views (e.g., synchronized selection, immediate rule checks) so users can move smoothly from exploration to planning. As a foundation for design, we will follow the design principles for a study planning assistant in higher education [4].

Basic Recommender System Component We leverage data collected through user interaction to implement a basic recommender component using established techniques. Recommendations are generated using established techniques such as content-based filtering [11, 23] (considering personal circumstances, interests, and competencies), collaborative filtering [7, 11, 3] (considering similarities with previous student study paths from prior student users), or a combination of both.

Integrating Recommendations into the Planning Workflow We embed recommendations directly into the graph–table workflow at key decision points (e.g., course selection, term placement, unmet requirements). Suggestions are contextual and briefly explained, with simple controls to accept, defer, or dismiss. Three types of recommendations are implemented:

- **Similar courses**: course alternatives with comparable topics.
- **Interest-matching courses** (content-based filtering): suggestions aligned with stated interests, or preferred competency areas.
- **Peer-informed choices** (collaborative filtering): courses frequently taken by prior students with similar plans or profiles.

The user study described above will also probe where and how these recommendations are most helpful.

User Study Following *ISO 9241-11* [1], we assess effectiveness, efficiency, and satisfaction in a task-based evaluation. Effectiveness and efficiency are captured through task performance data (e.g., task success, completion time, and errors). Satisfaction and perceived experience are measured using the *User Experience Questionnaire (UEQ* [2]) and the *Technology Acceptance Model (TAM)* [3] subscales for Perceived Usefulness and Perceived Ease of Use. We use a counterbalanced within-subjects design with two matched persona-based planning scenarios: participants complete one with the hybrid graph–table interface and one with a baseline (spreadsheet/paper). Additionally, participants perform a personal planning task to provide subjective qualitative feedback. The evaluation focuses primarily on the usability and user experience of the hybrid interface, while the recommender component is assessed only in terms of its perceived usefulness within the workflow.

4 State of the Art

Existing approaches to study planning address the problem from different angles but remain fragmented. Tools like STOPS [2], an early graph-based framework and prototype [19], and a recent elective-course visualization prototype [22] make prerequisite structures transparent and support exploration. These systems excel at surfacing dependencies, yet typically stop short of semester scheduling and workload orchestration. Complementing scheduling-oriented tools address timelines and advising: the Study Planning Assistant [13], individualized path planning [14], and an advising/progression dashboard [5]. CourseRank integrates search, ratings, and schedule building within a specific institutional context [6]. An Academic Course Planning Software System built for an Electrical and Communication program focuses on course selection and requirement checks in its local context [16].

Analytics approaches mine institutional logs to detect risks, compliance, and typical paths [20, 8, 25, 24]. Curriculum-mapping tools support alignment and coverage analyses [12, 1, 15]. Recent study-path analytics work shows how cohort pathways can inform quality assurance and planning support at program level [18]. These approaches deliver valuable institutional insight but are tightly coupled to LMS or CMS infrastructures and are often not directly embedded in student planners.

A rich body of work explores algorithms for recommending courses using collaborative filtering, content based filtering, or hybrid solutions [7, 26, 11, 3, 23], as well as optimization-based planners [21]. Most are evaluated offline or as algorithmic studies rather than as integrated planning tools. An exception is PlanGlow, an explainable, con-

[1] https://www.iso.org/standard/63500.html
[2] https://www.ueq-online.org/
[3] https://www.usersense.ch/wissensdatenbank/usability-metrics/technology-acceptance-model-tam

trollable LLM-driven study-planning system that foregrounds transparency in student-facing recommendations [10].

This thesis goes beyond the state of the art by combining the three strands, graph-based visualization, semester planning tables, and ERS, into a unified, student-facing tool. Unlike prior work, the system generates its own planning data during use, enabling process analytics independent of institutional infrastructures and providing real-time, personalized recommendations within the interface and planning workflow. The research question is therefore non-trivial: it asks not only how recommendations or analytics can be computed, but how they can be embedded directly into the planning workflow to improve real-world study decisions. We will use design principles for digital study-planning assistants to guide our development [4].

5 Context within the Master Program

The thesis aligns with the Software Engineering Master's program by addressing areas such as software engineering, human-computer interaction, and data science. Conceptually, it also connects to topics explored in the TU Wien courses Learning Analytics (VU) and Learning Technologies (SE), reflecting its emphasis on the educational domain.

References

[1] Susan Gardner Archambault and Jennifer Masunaga. "Curriculum Mapping as a Strategic Planning Tool". en. In: *Journal of Library Administration* 55.6 (2015), pp. 503–519. DOI: 10.1080/01930826.2015.1054770.

[2] Tapio Auvinen, Juha Paavola, and Juha Hartikainen. "STOPS: a graph-based study planning and curriculum development tool". en. In: *Koli Calling '14: 14th Koli Calling International Conference on Computing Education Research*. Association for Computing Machinery, 2014, pp. 25–34. DOI: 10.1145/2674683.2674689.

[3] Amer Al-Badarenah and Jamal Alsakran. "An Automated Recommender System for Course Selection". en. In: *International Journal of Advanced Computer Science and Applications* 7.3 (2016). DOI: 10.14569/IJACSA.2016.070323.

[4] Lena Bartel et al. "Design Principles for a Study Planning Assistant in Higher Education". In: *Conference on Human Information Interaction and Retrieval*. Association for Computing Machinery, 2024, pp. 243–253. DOI: 10.1145/3627508.3638327.

[5] Frederik Baucks et al. "Ein Dashboard für die Studienberatung: Technische Infrastruktur und Studienverlaufsplanung im Projekt KI". de. In: *Workshops der 21. Fachtagung Bildungstechnologien (DELFI)*. 2023. DOI: 10.18420/wsdelfi2023-53.

[6] Benjamin Bercovitz et al. "CourseRank: a social system for course planning". en. In: *2009 ACM SIGMOD International Conference on Management of data*. Association for Computing Machinery, 2009, pp. 1107–1110. DOI: 10.1145/1559845.1559994.

[7] Kiratijuta Bhumichitr et al. "Recommender Systems for university elective course recommendation". In: *14th International Joint Conference on Computer Science and Software Engineering (JCSSE)*. NakhonSiThammarat, Thailand: IEEE, 2017, pp. 1–5. DOI: 10.1109/JCSSE.2017.8025933.

[8] Rüdiger Buck-Emden and Franz-Dominik Dahmann. "Analyse von Studienverläufen mit Process-Mining-Techniken". de. In: *HMD Praxis der Wirtschaftsinformatik* 55.4 (2018), pp. 846–865. DOI: 10.1365/s40702-018-0410-1.

[9] Sorathan Chaturapruek et al. "How a data-driven course planning tool affects college students' GPA: evidence from two field experiments". en. In: *L@S '18: Fifth (2018) ACM Conference on Learning @ Scale*. Association for Computing Machinery, 2018, pp. 1–10. DOI: 10.1145/3231644.3231668.

[10] Jiwon Chun et al. "PlanGlow: Personalized Study Planning with an Explainable and Controllable LLM-Driven System". en. In: *L@S '25: Twelfth ACM Conference on Learning @ Scale*. Association for Computing Machinery, 2025, pp. 116–127. DOI: 10.1145/3698205.3729541.

[11] A. Esteban, A. Zafra, and C. Romero. "Helping university students to choose elective courses by using a hybrid multi-criteria recommendation system with genetic optimization". en. In: vol. 194. Knowledge-Based Systems, 2020. DOI: 10.1016/j.knosys.2019.105385.

[12] R. M. Harden. "Curriculum mapping: a tool for transparent and authentic teaching and learning". en. In: *Medical Teacher* 23.2 (2001), pp. 123–137. DOI: 10.1080/01421590120036547.

[13] Tobias Hirmer, Jana Etschmann, and Andreas Henrich. "Requirements and Prototypical Implementation of a Study Planning Assistant in CS Programs". In: *International Symposium on Educational Technology (ISET)*. Hong Kong, Hong Kong: IEEE, 2022, pp. 281–285. DOI: 10.1109/ISET55194.2022.00066.

[14] Sven Judel et al. "Supporting Individualized Study Paths Using an Interactive Study Planning Tool". en. In: *21. Fachtagung Bildungstechnologien (DELFI)*. 2023. DOI: https://doi.org/10.18420/delfi2023-36.

[15] Naim Kapucu. "Competency-based Curriculum Mapping as a Tool for Continuous Improvement for Master of Public Administration (MPA) Programs". In: *International Journal of Public Administration* 40.11 (2017), pp. 968–978. DOI: 10.1080/01900692.2016.1229677.

[16] Mohammad Shakeel Laghari et al. "Academic Course Planning Software System at EECE Department". en. In: *12th International Conference on Software and Computer Applications*. Kuantan Malaysia: Association for Computing Machinery, 2023, pp. 97–104. DOI: 10.1145/3587828.3587844.

[17] Sara Morsy and George Karypis. "A Study on Curriculum Planning and Its Relationship with Graduation GPA and Time To Degree". en. In: *The 9th International Learning Analytics & Knowledge Conference*. Association for Computing Machinery, 2019, pp. 26–35. DOI: 10.1145/3303772.3303783.

[18] Rene Roepke, Sven Judel, and Ulrik Schroeder. "Study path analyses for quality assurance and support of study planning: Approaches and advancements in the AIStudyBuddy project". en. In: *Informatik Spektrum* 47 (2024), pp. 97–104. DOI: 10.1007/s00287-024-01574-y.

[19] Raita Rollande and Janis Grundspenkis. "Graph based framework and its implemented prototype for personalized study planning". In: *2013 Second International Conference on E-Learning and E-Technologies in Education (ICEEE)*. IEEE, 2013, pp. 137–142. DOI: 10.1109/ICeLeTE.2013.6644362.

[20] Jurgen Schulte et al. "Large scale predictive process mining and analytics of university degree course data". en. In: *LAK '17: 7th International Learning Analytics and Knowledge Conference*. Association for Computing Machinery, 2017, pp. 538–539. DOI: 10.1145/3027385.3029446.

[21] Alangkarn Srisamutr, Thitiporn Raruaysong, and Vacharapat Mettanant. "A Course Planning Application for Undergraduate Students Using Genetic Algorithm". In: *2018 Seventh ICT International Student Project Conference (ICT-ISPC)*. Nakhonpathom: IEEE, 2018, pp. 1–5. DOI: 10.1109/ICT-ISPC.2018.8523980.

[22] Marie Trippel and René Röpke. "Developing a Graph-based Visualization of Elective Courses to Support Course Selection in Higher Education". en. In: *23. Fachtagung Bildungstechnologien (DELFI)*. 2025. DOI: 10.18420/delfi2025_09.

[23] Thi-Hai-Yen Vuong et al. "A Personalized Course Recommender System for Undergraduate Students". In: *International Journal of Learning and Teaching* (2019), pp. 181–190. DOI: 10.18178/ijlt.5.3.181-190.

[24] Miriam Wagner et al. "A Combined Approach of Process Mining and Rule-Based AI for Study Planning and Monitoring in Higher Education". en. In: *Process Mining Workshops*. Vol. 468. Springer Nature Switzerland, 2023, pp. 513–525. DOI: 10.1007/978-3-031-27815-0_37.

[25] Ren Wang and Osmar R. Zaïane. "Discovering Process in Curriculum Data to Provide Recommendation." In: *8th International Conference on Educational Data Mining*. 2015, pp. 580–581.

[26] Chris Wong et al. "Sequence Based Course Recommender for Personalized Curriculum Planning". In: *Artificial Intelligence in Education*. Vol. 10948. Springer International Publishing, 2018, pp. 531–534. DOI: 10.1007/978-3-319-93846-2_100.